

PLABABLE

GEMS

VERSION 1.0

CRITICAL CARE



Tricyclic Antidepressant Overdose

Presentation

- Dilated pupils
- Dry mouth and skin
- Hypotension
- Urinary retention
- Tachycardia

Investigation

- **ECG:** Widened QRS and broad complex tachycardia
- **ABG:** Metabolic acidosis

Management

- IV Sodium bicarbonate (50 ml of 8.4%)
- IV fluids (NS)
- ECG monitoring



Paracetamol Overdose

Presentation

- Right upper quadrant pain
- Nausea and vomiting
- Liver failure (NAPQI toxicity)
 - Jaundice
 - Encephalopathy
 - Lactic acidosis

Investigation

- Paracetamol (4 hrs post intake)
- ↑ ALT
- ↑ PT/INR
- Hypoglycemia

Management

- **Activated charcoal** if presented <1 hr after intake
- **IV N-acetylcysteine** if at 4th hr after intake, paracetamol value is above the treatment line
- **IV N-acetylcysteine** if staggered dosing or ingestion of unknown amount irrespective of paracetamol value
- **Liver transplantation** in case of liver failure

Refeeding Syndrome

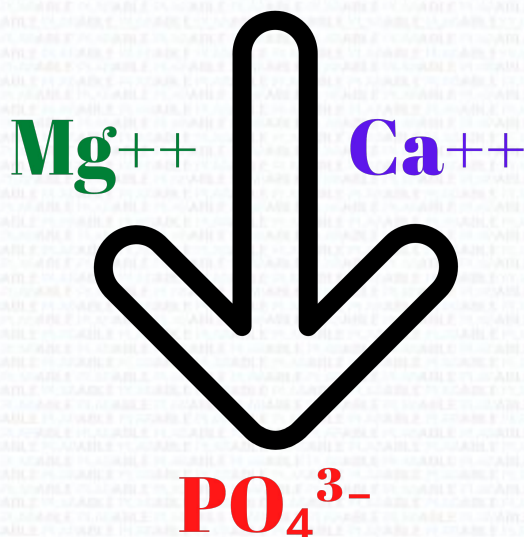
Metabolic disturbances occurring after reinstitution of carbohydrate to starved patients

Presentation

- Hypophosphatemia
- Hypomagnesaemia
- Hypocalcaemia

Management

- Correction of electrolyte imbalance
- IV fluids
- **Prevention:** Feeds should be given slowly with electrolyte supplementation



Refeeding Syndrome

Brain trainer:

A 21 year old woman with a BMI of 12 was admitted to the medical ward for feeding through a nasogastric tube. What is electrolyte abnormality should be expected?

➔ **Hypophosphataemia**

Subarachnoid Haemorrhage

MC cause: Bleeding from a **berry aneurysm**

Presentation

- Sudden and severe headache (worst headache of life or **thunderclap** headache)
- Neck stiffness or pain

Risk factors

- Hypertension
- Smoking
- Polycystic kidney disease (↑ Berry aneurysm)



**Thunderclap
Headache**



**Berry
Aneurysm**

Subarachnoid Haemorrhage

Investigation

- **CT head** (first line) - hyperdense appearance of blood in the basal cisterns
- Cerebral angiography (gold standard)
- If CT is normal but there is still a clinical suspicion: lumbar puncture after 12 hrs - **xanthochromia**

Treatment

- **Nimodipine** to prevent cerebral vasospasm
- Platinum coiling to stop bleeding



Subarachnoid Haemorrhage

Brain trainer:

A 35 year old man attends the ED with a severe headache and vomiting for a day. He has neck stiffness and photophobia. Head CT is shown below.



What is the most appropriate management?

→ **Nimodipine**

Subarachnoid Haemorrhage

Brain trainer:

A 32 year old man presents with “the worst headache of his life”. He has photophobia, neck stiffness and has vomited multiple times. CT scan shows no evidence of an intracranial bleed. What is the most appropriate next step?

➔ Lumbar puncture

Extradural (Epidural) Haematoma

Collection of blood between the dura and the skull

Presentation

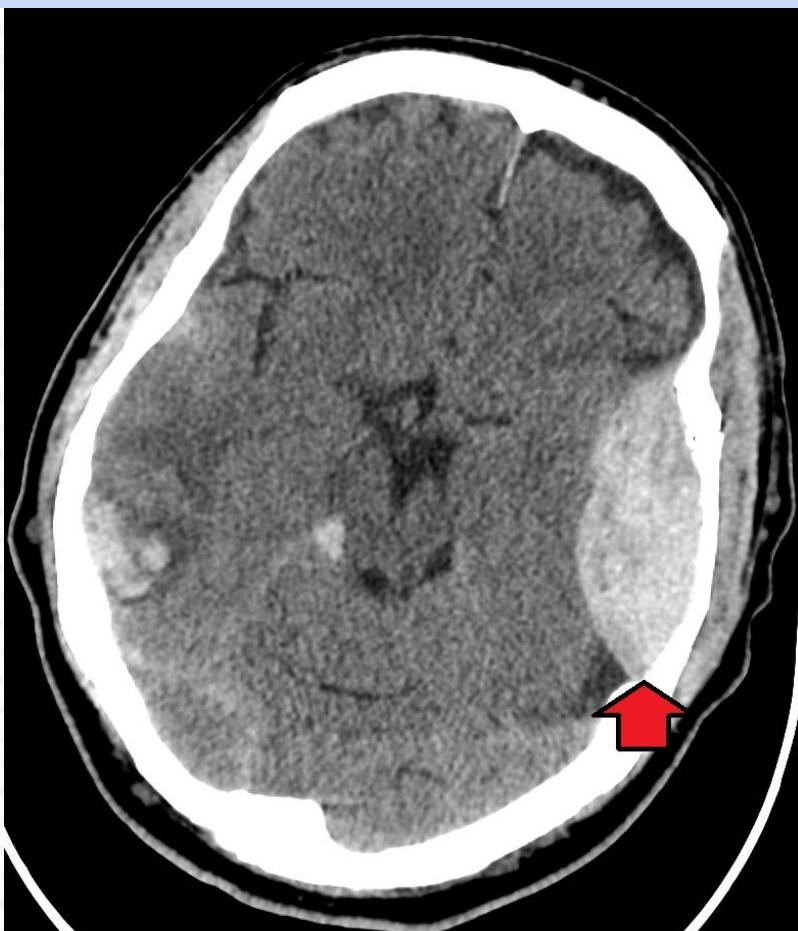
- Skull fracture → **Middle meningeal artery** rupture
- Loss of consciousness immediately → **Lucid interval** → Deterioration of condition
- Blown pupil or pupillary asymmetry due to ↑ ICP

Investigation

- **CT head:** Bi-convex or lentiform shaped haematoma

Management

- **Craniotomy** or burr hole (big haematoma)
- **Conservative** (small haematoma)



Subdural Haematoma

Collection of blood between dura and arachnoid mater

Presentation

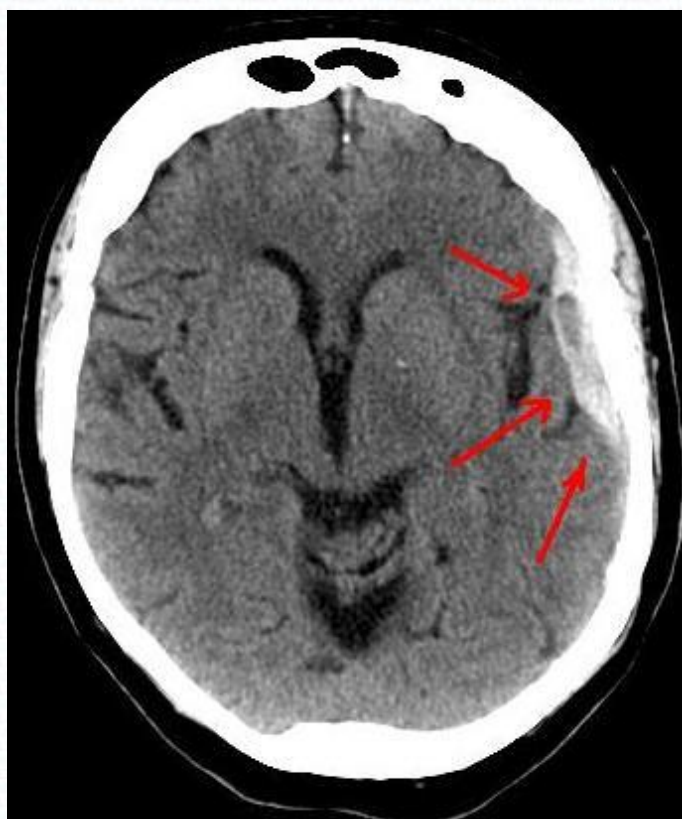
- Associated with **blunt head injury**
- **Chronic SDH** is seen in **elderly**
 - Gradually evolving focal neurological deficits
 - Speech difficulties and
 - Drowsiness
 - H/O Anticoagulation
- ↑ ICP

Investigation

- **CT head:** crescent-shaped haematoma

Management

- Craniotomy or burr hole (big haematoma)
- Conservative (small haematoma)



Acute Chest Syndrome

Vaso-occlusive crisis due to **sickle-cell disease**

Presentation

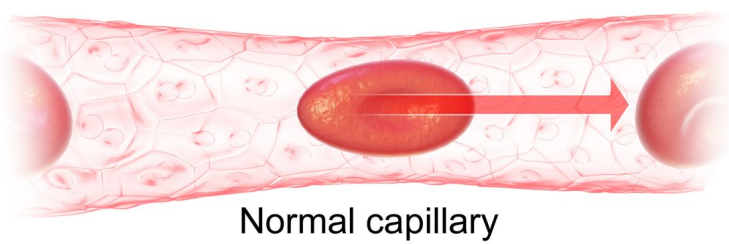
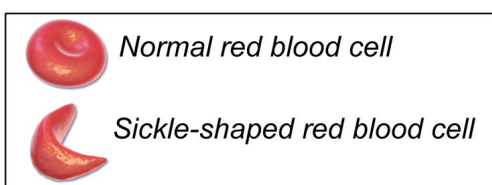
- Chest pain
- Shortness of breath
- Cough
- Triggered by infection (Pneumonia)
- H/o sickle-cell disease

Investigation

- Chest X-ray
- Monitoring oxygen saturation

Management

- Oxygen supplementation
- Analgesia
- IV fluids
- Antibiotics to treat pneumonia



Massive Pulmonary Embolism

Presentation

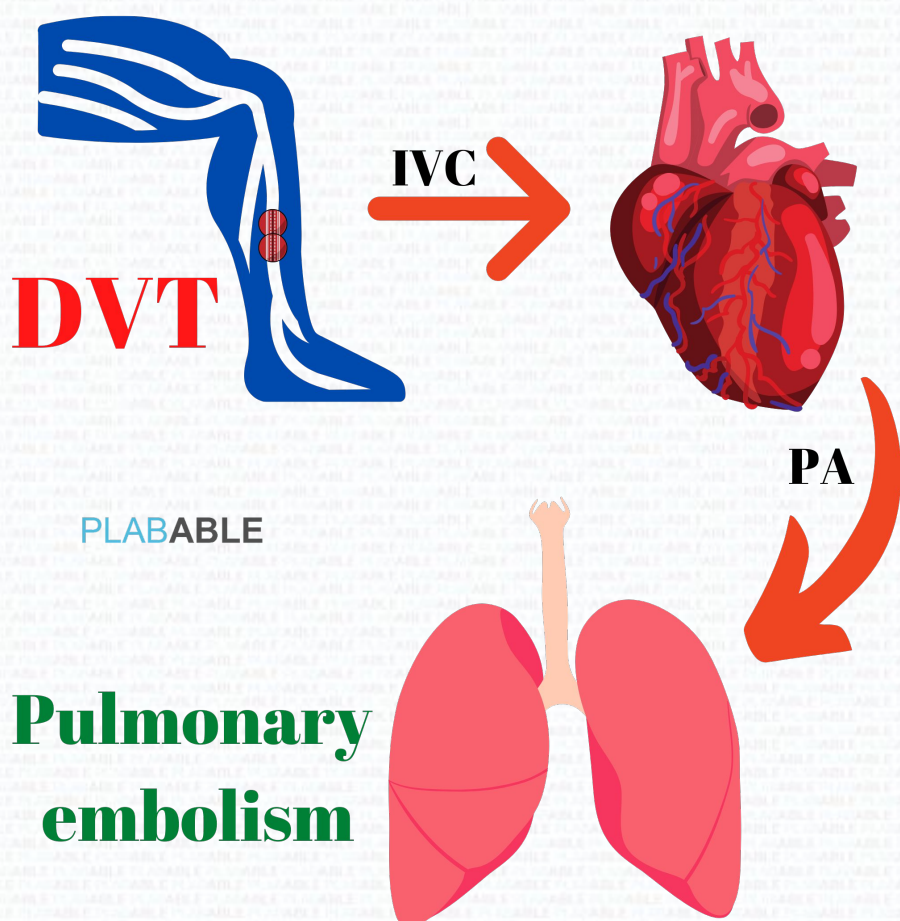
- Breathlessness
- Haemodynamic instability
- Leg swelling (DVT)

Investigation

- CT pulmonary angiography (CTPA)

Management

- Thrombolysis
- Surgical embolectomy



Blunt Abdominal Trauma

Diaphragm rupture

- RTA
- Chest and abdominal pain
- Diminished breath sounds on the affected side
- **X-ray:** Gas bubbles in the pleural cavity

Splenic rupture

- RTA
- Abdominal pain
- Tenderness in the epigastrium and left flank
- Hypotension
- FAST scan abdomen
- Splenectomy (severe injury)



Diaphragm Rupture

Brain trainer:

What is the most accurate method for checking the correct placement of a nasogastric tube?

→ **Chest X-ray**

Splenic Rupture

Brain trainer:

What is the most commonly injured organ ?

→ **The spleen**

Splenic Rupture

Brain trainer:

What is the imaging modality of choice with a patient suspected of splenic rupture?

→ **FAST (sonography - when clinically unstable)**

→ **CT (when clinically stable)**

Renal Trauma

Brain trainer:

A man falls from a ladder and receives a bump on his flank. He has haematuria and is hypotensive. What is the most appropriate management?

1. IV. fluids until stable

**2. Urgent CT abdomen
(FAST if CT not an option)**

Status Asthmaticus

Presentation

- Acute breathlessness
- H/o asthma
- ↓ SpO₂
- Poor respiratory effort

Management

Salbutamol nebulisation + oxygen



Ipratropium bromide nebulisation



Intravenous hydrocortisone



**Intravenous salbutamol and
aminophylline**



Intravenous magnesium sulphate



Intubation and ventilation

Status Asthmaticus

Brain trainer:

A 6 year old child with breathlessness for the past 12 hours presents to the ED. He is drowsy with poor respiratory effort and has a silent chest. He is started on oxygen.

What is the most appropriate management?

➔ **Intubation and ventilation**

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